

SpaceX Falcon 9 First Stage Landing Prediction

- Data Science Capstone Project
- API, EDA, SQL, geospatial analysis, dashboard, and machine learning
- GitHub: github.com/madao009papa/github-getting-started-lab-20260811

Why this problem matters

- Reusable Falcon 9 boosters can reduce launch cost.
- Predicting landing success supports operational and bidding decisions.
- Target: binary landing class - successful or not successful.

Data and workflow

- SpaceX API: launch, rocket, payload, launchpad, and core metadata.
- Web scraping: Falcon launch history table.
- Cleaning: Falcon 9 filtering, missing payload imputation, feature creation.
- Analysis: EDA, SQL, Folium, Plotly Dash, and four classifiers.

Evidence from analysis

- Four launch sites were identified.
- NASA (CRS) payload total: 45,596 kg.
- Average F9 v1.1 payload: 2,534.67 kg.
- First successful ground-pad landing: 2015-12-22.
- 2015 drone-ship failures: two records.

Model comparison

- Logistic Regression: test accuracy 0.833.
- Support Vector Machine: test accuracy 0.833.
- Decision Tree: test accuracy 0.667.
- K Nearest Neighbors: test accuracy 0.833.
- Recommended baseline: Logistic Regression for stability and interpretability.

Submission status

- GitHub repository created and populated with 10 capstone artifacts.
- README maps evaluation criteria 1.1-1.15 to the artifacts.
- Final report PDF and presentation PDF are included with this submission.
- Final action: upload the final PDF to Mark.